

GREENBRIER VALLEY, WV, USA

WMO: 724127

Lat: 37.867N

Lon: 80.400W

Elev: 702

StdP: 93.17

Time Zone: -5.00 (NAE)

Period: 00-19

WBAN: 53801

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.7	-11.3	-19.7	0.7	-10.9	-17.1	0.9	-8.6	8.3	1.8	7.5	0.5	2.2	300	0.429

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	30.6	20.8	29.0	20.4	27.7	20.0	23.0	27.1	22.3	26.4	21.6	25.6	2.6	270

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
22.1	18.3	25.1	21.2	17.3	24.4	20.3	16.3	23.6	72.0	27.3	69.1	26.2	66.6	25.4	35.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.2	5.8	5.1	-17.5	32.6	3.2	4.3	-19.9	35.7	-21.7	38.2	-23.5	40.7	-25.9	43.8		
(5)			-17.9	24.4	3.2	2.9	-20.1	26.4	-22.0	28.1	-23.8	29.8	-26.1	31.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.8	-0.9	0.8	5.1	11.2	15.5	19.4	21.2	20.8	17.7	11.5	5.9	1.2
	DBStd	9.09	6.07	5.52	5.76	4.96	4.38	2.99	2.68	2.70	3.67	4.74	4.83	5.36
	HDD10.0	1284	339	263	172	44	8	0	0	0	2	39	139	278
	HDD18.3	3033	596	492	409	215	106	24	7	9	54	217	372	532
	CDD10.0	1592	2	4	21	82	180	281	347	336	231	86	17	4
	CDD18.3	298	0	0	0	2	19	55	96	87	33	4	0	0
	CDH23.3	2326	0	0	5	59	192	426	697	622	281	43	1	0
	CDH26.7	526	0	0	0	7	28	84	182	153	66	7	0	0
Wind	WSAvg	1.7	2.2	2.2	2.2	2.2	1.7	1.5	1.3	1.1	1.2	1.5	1.6	2.0
Precipitation	PrecAvg	971	71	69	86	83	97	86	103	86	74	66	69	74
	PrecMax	1451	175	168	198	181	204	206	231	199	174	159	186	156
	PrecMin	670	9	14	11	13	22	21	46	23	1	1	8	13
	PrecStd	166	34	31	42	37	46	46	43	43	45	39	34	34
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.1	18.9	23.7	27.6	29.0	31.1	32.5	32.3	31.2	27.2	22.1	18.0
		MCWB	13.1	13.5	14.0	14.6	18.9	21.0	21.3	22.0	19.3	18.5	15.7	15.0
	2%	DB	13.3	15.2	20.7	25.0	27.3	28.9	30.2	30.0	28.7	24.0	18.9	14.6
		MCWB	10.9	11.0	13.2	15.1	18.3	20.4	21.2	20.7	19.4	16.9	13.3	12.2
	5%	DB	11.0	12.4	17.5	22.7	25.8	27.5	28.8	28.0	26.9	22.2	17.0	12.1
		MCWB	9.0	8.4	11.4	14.0	17.8	19.7	20.9	20.7	19.2	16.1	12.6	9.8
	10%	DB	7.8	9.8	15.0	20.9	23.7	26.1	27.4	27.2	24.5	19.8	14.1	9.4
		MCWB	6.0	6.6	10.1	13.6	17.2	19.4	20.8	20.4	18.3	15.6	10.8	7.6
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.1	15.1	15.8	18.5	20.8	22.7	23.9	24.1	22.3	20.5	17.0	15.8
		MCDB	15.9	18.0	21.3	24.1	26.2	27.5	27.9	28.6	26.7	25.0	19.1	17.9
	2%	WB	11.8	12.1	14.1	17.0	20.0	21.8	23.1	23.0	21.2	19.2	15.1	12.8
		MCDB	13.0	14.2	18.5	21.6	24.9	26.3	27.2	27.0	25.4	22.5	18.1	14.3
	5%	WB	8.9	9.1	12.8	15.7	19.1	21.2	22.4	22.2	20.4	17.6	13.4	10.2
		MCDB	10.4	11.6	16.5	20.0	23.6	25.5	26.4	26.0	24.3	19.8	15.6	11.6
	10%	WB	6.2	6.5	10.9	14.3	18.1	20.4	21.7	21.3	19.4	15.9	11.6	7.7
		MCDB	7.8	9.0	13.8	18.7	21.9	24.6	25.6	25.0	23.1	18.8	13.4	8.9
Mean Daily Temperature Range	5% DB	MDBR	9.1	9.8	11.2	12.9	11.8	11.3	10.7	10.9	11.6	11.8	11.4	8.9
		MCDBR	12.4	14.3	15.6	16.8	14.5	13.0	12.6	12.7	14.5	15.2	15.4	12.3
		MCWBR	9.7	10.0	9.0	8.2	6.7	5.8	5.0	5.3	6.3	8.2	10.2	9.4
	5% WB	MCDBR	11.0	12.2	13.6	12.9	12.3	11.3	10.8	10.7	11.5	11.4	11.9	11.2
		MCWBR	9.5	9.8	8.8	7.3	6.5	5.9	5.7	5.5	5.8	7.0	9.1	9.6
Clear-Sky Solar Irradiance	taub	0.292	0.307	0.331	0.370	0.409	0.433	0.462	0.468	0.369	0.342	0.307	0.290	
	taud	2.496	2.440	2.432	2.351	2.244	2.206	2.153	2.109	2.384	2.459	2.529	2.532	
	Ebn at Noon	900	926	932	908	874	849	821	806	879	873	874	879	
	Edh at Noon	79	95	106	122	138	143	150	153	110	92	76	71	
All-Sky Solar Radiation	RadAvg	2.00	2.73	3.79	4.90	5.43	5.98	5.65	5.20	4.36	3.32	2.42	1.76	
	RadStd	0.20	0.35	0.26	0.39	0.47	0.40	0.34	0.27	0.46	0.41	0.26	0.17	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (32 neighbors)	+0.40	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+127	+63

Nomenclature: See separate page